30 June 1955
SUPERSEDING
NAVORD OS 1364 B
16 August 1951

MILITARY SPECIFICATION
GROMMETS, FIBER
(FOR PROJECTILE ROTATING BANDS)

1. SCOPE

1.1 Scope.- This specification covers certain sizes and types of fiber grommets for the protection of projectile rotating bands.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-F-3192 Fiberboard, Flame-Resistant (For Projectile Grommets)

MIL-C-15159 Compound, Preservative; Fire-Water-Mildew,- and Weather-Resistant (For Canvas)

MIL-P-16594 Projectiles, Bombs, Rockets and Guided Missile Warheads, Inert Parts, Manufacture of

MIL-C-17516 Copper-Silicon Alloy Bars, Plates, Rods, Sheets, Strips, Flat Wire, Wire, Forgings, and Structural and Special Shaped Sections.

NAVY DEPARTMENT

General Specifications for Inspection of Material, Appendix II (Metals)

STANDARDS

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 Marking of Shipments

DRAWINGS

BUREAU OF ORDNANCE

330051 Fiber Grommets (For Projectile Rotating Bands)

PUBLICATIONS

NAVY ORDNANCE PAMPHLETS

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OP400 General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Material

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- In addition to the requirements specified herein the requirements of Specification MIL-P-16594 shall apply.

3.1.1 Pre-production lot.- Unless otherwise specified the submission of a pre-production lot and the satisfactory compliance of this lot with the requirements of this specification shall be required before general production on the contract is approved. These grommets shall be made by the same processes the contractor intends to use in the fulfillment of the contract.

3.2 Material.- The body and swells of the grommets shall be made of fiberboard in accordance with Specification MIL-F-3192 and when specified on the applicable drawing shall be reinforced on the outside with metal bands. The grain of the fiber, if any, shall be in the longest direction of the piece.

3.2.1 Metal parts.- All metal parts shall be of steel, made corrosion resistant by zinc plating or by a suitable coating, or shall be of copper-alloy in accordance with Specification MIL-C-17516. In either case the metal parts shall be required to pass the salt spray tests of 4.3.1. The metal parts shall not crack, break, or show other indications of weakness when subjected to the conditions of use.

3.2.1.1 Eyelets shall be of adequate size to permit quick assembly of the laces and shall be properly assembled to the body of the grommet such that each assembled eyelet shall withstand a minimum pull of 70 pounds for 5 and 6 inch and 120 pounds for 8 inch and larger grommets without indication of weakness in the eyelet or surrounding fiber.

3.2.1.2 Fasteners for securing the swells to the body may be either rivets or staples. The fasteners shall pass through all the pieces to be fastened and shall be adequately backed and clinched on both sides to prevent pulling out. Fasteners shall not protrude more than 0".020 beyond the inner surface of the swells. Fasteners shall be placed to each side of the centerline of the rear swell as shown on the applicable drawing.

3.2.2 Laces.- The material of the laces shall be mildew proof or properly treated to conform to the mildew resistance requirements of Specification MIL-C-15159. The ends of the laces shall be tipped to facilitate assembly with the grommets. The breaking strength of the laces shall not be less than 70 pounds for 5 and 6 inch grommets nor less than 120 pounds for 8 inch and larger grommets. Unless otherwise specified laces shall not be assembled with the grommets but shall accompany each package of grommets and the number furnished shall be 110 percent of the number of grommets.

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3.3 Shape.- Grommets shall be delivered in the form of open rings and shall be flexible enough to allow the ends of the grommet to spread to make a gap greater than the outside diameter of the rotating band at assembly. No appreciable cracking shall occur in the fiber when so opened. Grommets shall be capable of rapid assembly, fastening, and removal from projectiles by hand without the use of special tools. The body and swells may be made in one piece or fabricated with laminated swells as shown, provided the finished product meets the dimensional and functional requirements prescribed by the drawing and specifications.

3.4 Fit.- Grommets shall make a snug fit when assembled on projectiles. The diameters of the surfaces on the projectiles contacting the grommets, shown on the drawing for the applicable grommet, and the space specified as a gap at the ends of the grommet shall determine the contacting peripheries of the grommet. All surfaces shall be free from burrs, sharp edges, etc., that could injure the bare hand.

3.5 Marking.- Each grommet shall be legibly and indelibly marked with the caliber, drawing number, assembly number, and the drawing revision letter in effect at the time of manufacture and an arrow, at least one inch long, pointing toward the front edge. Marking may be by painting, stamping or stencilling.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling.- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government Inspection will be in accordance with the provisions of MIL-STD-105.

4.1.1 Inspection lot.- As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot," i.e. a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" shall be as determined by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Inspection.- The general inspection procedures shall be in accordance with General Instructions For the Design, Manufacture, and Inspection of Naval Ordnance Equipment, OP400; Government Inspection shall be in accordance with the provisions of General Procedure for Government Inspection and Contractor's Inspection shall be in accordance with Inspection by Contractor, as contained in OP400.

4.3 Tests at contractor's plant.- In addition to the tests described herein, the Inspector may require other tests, measurements, etc., necessary to insure that the grommets are in accordance with all requirements.

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4.3.1 Salt spray test.- Representative specimens of the metal parts shall be subjected to the twenty-percent salt spray (fog) test in accordance with General Specifications for Inspection of Material, Appendix II, Metals, Part A. No rust spots shall be formed, except at ends if not coated, after twenty-four hours exposure.

4.3.2 Eyelet pull test.- Assembled eyelets shall be subjected to the specified loading applied on a loop of wire or cable passing through the eyelet.

4.3.3. Lace breaking strength.- Laces shall be subjected to the specified loading.

4.4 Pre-production lot test.- Grommets in the pre-production lot will be subjected to complete inspection and to all tests considered necessary by the Inspector to assure compliance with the requirements of the drawing and this specification. The results of this examination will determine whether the contractor shall proceed under the contract. Unless otherwise specified a minimum of fifty grommets shall comprise a pre-production lot.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Unless otherwise specified commercial packages meeting requirements of common carriers are acceptable. No more than one size of grommet shall be packed in a single container.

5.2 Marking.- Each package shall be marked with the following: Grommets, caliber, drawing and assembly number, the quantity contained therein, and the number of the contract or order. Standard MIL-STD-129 applies to general marking requirements.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

a) Title, number, and date of this specification.
b) Drawing number and date of applicable drawing.

6.2 Use.- Grommets covered by this specification are assembled over the rotating bands of projectiles to protect them from damage during handling, shipment, and stowage.

6.3 Precedence.- All correspondence, instructions, or practices that are in conflict with and are dated or in use prior to the latest revision of this specification are hereby cancelled.

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Patent Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

FORM APR 63 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE C-6279

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PLATE NO. 15419 (BACK)

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